

# Bayesian Hierarchical Model for PM<sub>2.5</sub> levels in Italy

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# 1 Introduction

Air pollution represents a major environmental and public health concern. Specifically, fine particulate matter  $\text{PM}_{2.5}$  is among the most harmful pollutants due to its ability to penetrate deep into the respiratory system and its strong association with cardiovascular and pulmonary diseases. Reliable statistical modeling of  $\text{PM}_{2.5}$  concentrations is therefore important for understanding pollution patterns and for assessing the persistence of environmental exposure across different locations.

A key challenge in modeling  $\text{PM}_{2.5}$  concentrations lies in the presence of substantial heterogeneity across monitoring stations and irregular temporal coverage. Stations differ significantly in terms of geographical characteristics, local emission sources, and environmental conditions, while observational data often exhibit an unbalanced panel structure, with varying numbers of observations across stations and missing time periods. As a result, direct comparisons across stations may be misleading, since observed differences can reflect both genuine site-specific pollution patterns and differences in the amount and timing of available data.

In this context, Bayesian hierarchical models provide a natural and flexible framework for analyzing data. By modeling variation at multiple levels, they make it possible to account for systematic differences across observational units, while still leveraging information from the entire dataset. This is particularly useful when some stations have limited observations, as it avoids relying exclusively on sparse local data and allows inference to be stabilized through partial pooling.

Building on these considerations, this paper develops a Bayesian hierarchical model for daily  $\text{PM}_{2.5}$  concentrations observed across monitoring stations in Italy during 2025. Because the model includes a non-conjugate scale component, posterior inference is carried out using Markov Chain Monte Carlo (MCMC) methods, specifically a Metropolis-within-Gibbs sampler combining conjugate Gibbs updates with a Metropolis-Hastings step. The analysis focuses on model specification, MCMC implementation, convergence diagnostics, sensitivity analysis, and posterior predictive checks.

## 2 Data

The dataset comprises daily  $\text{PM}_{2.5}$  concentration measurements collected across monitoring stations in Italy during 2025. The raw data were sourced from the European Environment Agency (EEA) Air Quality portal, specifically via the EEA Air Quality e-Reporting, with metadata and station specifications validated through the EEA Air Quality Viewer.<sup>1</sup> After preprocessing and quality control, the final dataset contains 31,349 observations distributed across 116 stations. Since the subsequent model is specified for a continuous, approximately Gaussian outcome, the distributional properties of  $\text{PM}_{2.5}$  concentrations were inspected before estimation.

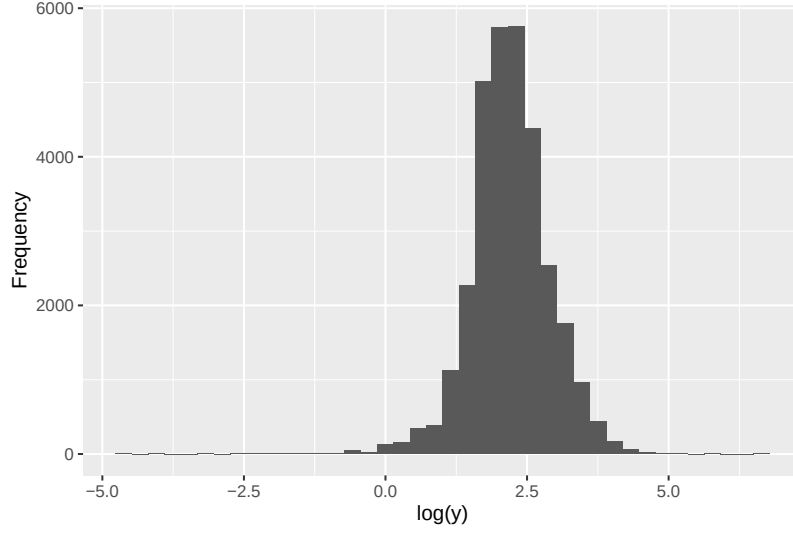
The empirical distribution of the raw outcome variable was highly right-skewed, with a skewness of approximately 17.1, primarily driven by extreme pollution events in the upper tail. Therefore, to stabilize variance and improve compatibility with the Gaussian likelihood of the proposed model, the outcome variable was transformed as

$$y_{it} = \log(\text{PM}_{2.5,it}),$$

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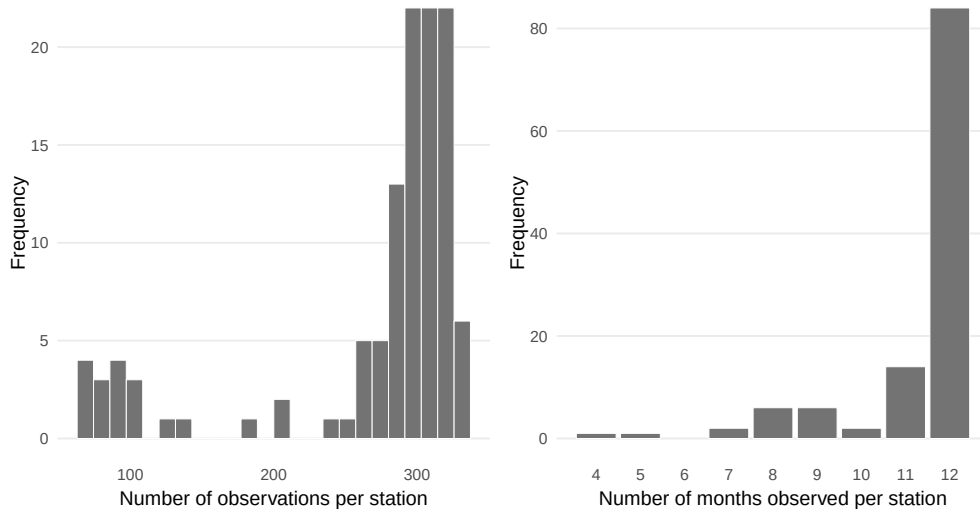
<sup>1</sup>EEA Air Quality e-Reporting: <https://eadmz1-downloads-webapp.azurewebsites.net/>. EEA Air Quality Viewer: [https://discomap.eea.europa.eu/App/AQViewer/index.html?fqn=Airquality\\_Dissemination.b2g.measurements](https://discomap.eea.europa.eu/App/AQViewer/index.html?fqn=Airquality_Dissemination.b2g.measurements).

where  $y_{it}$  represents the transformed concentration for station  $i$  at time  $t$ . This transformation substantially reduced skewness to approximately -0.29, yielding a more symmetric distribution, as shown in Figure 1, and making the likelihood specification a reasonable modeling choice.



**Figure 1:** Distribution of the log-transformed PM<sub>2.5</sub> concentrations.

As is often the case in environmental monitoring networks, the panel resulting from the preprocessing stage is markedly unbalanced. The number of observations per station ranges from 69 to 332, with a median of approximately 299. Temporal coverage is also uneven across stations. While most stations provide observations for most of the year, 32 stations lack at least one full month of data, and for a few stations coverage drops to only 4 months. This heterogeneity is summarized in Figure 2, which reports both the distribution of the number of observations per station and the distribution of the number of months observed per station.



**Figure 2:** Data availability by station.

For the subsequent analysis, the cleaned dataset was restricted to the variables used in the empirical specification: the log-transformed PM<sub>2.5</sub> concentration, the station identifier, the month, and the

day of the week. Observations with missing values in any of these variables were excluded, resulting in a complete-case sample. The variables `station_id`, `month`, and `day_of_week` were then treated as categorical factors.

### 3 Model

The structure of the dataset naturally motivates a multilevel modeling approach to account for unobserved heterogeneity across monitoring stations. The Bayesian hierarchical specification enables partial pooling and improves estimation stability in the presence of uneven data availability.

The covariates included in the model correspond to calendar effects. In particular, the fixed-effects design matrix is constructed as

$$X = \text{model.matrix}(\sim \text{month} + \text{day\_of\_week}),$$

so that the intercept is included by default together with monthly and day-of-week indicators. The resulting parametrization corresponds to a treatment coding scheme, in which the intercept represents the expected log  $\text{PM}_{2.5}$  level for the reference categories (January and Monday), and the remaining coefficients measure deviations from these baselines. Station-level heterogeneity is not included among the fixed effects, but is instead modeled through station-specific random intercepts.

The likelihood is specified as:

$$y_{it} \mid \beta, \alpha, \sigma^2 \sim \mathcal{N}(x_{it}^\top \beta + \alpha_i, \sigma^2),$$

where  $i$  indexes monitoring stations and  $t$  indexes time. The vector  $x_{it}$  contains the calendar covariates,  $\beta$  is the vector of regression coefficients, and  $\alpha_i$  is the station-specific intercept capturing unobserved heterogeneity.

Equivalently, the model can be written as

$$y_{it} = x_{it}^\top \beta + \alpha_i + \varepsilon_{it}, \quad \text{with } \varepsilon_{it} \sim \mathcal{N}(0, \sigma^2).$$

To model cross-station variability, the random effects are specified hierarchically as

$$\alpha_i \mid \tau \sim \mathcal{N}(0, \tau^2), \quad i = 1, \dots, J,$$

where  $\tau^2$  represents the variance of baseline pollution levels across stations. The zero mean centers the station effects around the overall baseline level captured by the intercept and calendar covariates, so that  $\alpha_i$  represents a station-specific deviation. This specification induces partial pooling, shrinking estimates for sparsely observed stations toward the overall population mean.

#### 3.1 Prior specification

Weakly informative priors are assigned to all parameters to regularize estimation while allowing the likelihood to dominate posterior inference.

For the regression coefficients:

$$\beta \sim \mathcal{N}(b_0, B_0),$$

where  $B_0 = cI_p$  is a diagonal covariance matrix. This specification assumes prior independence across coefficients and imposes mild shrinkage toward zero, helping to stabilize estimation in the presence of potentially correlated temporal indicators.

The observation variance is assigned the prior

$$\sigma^2 \sim \text{Inverse-Gamma}(a_\sigma, b_\sigma),$$

where the inverse-gamma distribution is parameterized in terms of shape and scale. This prior provides conjugacy and simplifies posterior computation, while remaining weakly informative for appropriate choices of hyperparameters.

The group-level scale parameter  $\tau$  is assigned a half-Cauchy prior,

$$\tau \sim \text{Half-Cauchy}(0, A_\tau).$$

This choice provides weakly informative regularization for the scale of the random effects, allowing for substantial between-station variability while still shrinking implausibly large values. The heavy-tailed nature of the half-Cauchy distribution ensures that large values of  $\tau$  remain a priori possible, avoiding overly restrictive assumptions on the degree of between-station heterogeneity.

### 3.2 Posterior structure and MCMC estimation

Combining the likelihood with the prior distributions, the joint posterior density is proportional to

$$p(\beta, \alpha, \sigma^2, \tau \mid y) \propto \left[ \prod_{i,t} \mathcal{N}(y_{it} \mid x_{it}^\top \beta + \alpha_i, \sigma^2) \right] \left[ \prod_{i=1}^J \mathcal{N}(\alpha_i \mid 0, \tau^2) \right] \mathcal{N}(\beta \mid b_0, B_0) \cdot \text{IG}(\sigma^2 \mid a_\sigma, b_\sigma) \cdot \pi(\tau).$$

Although the joint posterior does not admit a closed-form expression, most full conditional distributions are available in closed form, while the scale parameter  $\tau$  does not admit a conjugate full conditional distribution under the half-Cauchy prior and therefore requires a non-standard update.

In particular, conditional on  $\sigma^2$  and  $\tau$ , the regression coefficients  $\beta$  follow a multivariate normal distribution:

$$\beta \mid \cdot \sim \mathcal{N}(\tilde{b}, \tilde{B}),$$

where the posterior mean and covariance matrix are given by

$$\tilde{B} = \left( B_0^{-1} + \frac{1}{\sigma^2} X^\top X \right)^{-1}, \quad \tilde{b} = \tilde{B} \left( B_0^{-1} b_0 + \frac{1}{\sigma^2} X^\top (y - \alpha) \right),$$

with  $y$  denoting the the  $N$ -dimensional stacked outcome vector and  $\alpha$  denoting the vector of station-specific effects replicated at the observation level.

Similarly, each station-specific effect admits a Gaussian full conditional:

$$\alpha_i \mid \cdot \sim \mathcal{N}(\tilde{a}_i, \tilde{V}_i),$$

where

$$\tilde{V}_i = \left( \frac{n_i}{\sigma^2} + \frac{1}{\tau^2} \right)^{-1}, \quad \tilde{a}_i = \tilde{V}_i \cdot \frac{1}{\sigma^2} \sum_{t \in \mathcal{T}_i} (y_{it} - x_{it}^\top \beta),$$

with  $n_i$  denoting the number of observations available for station  $i$ , and  $\mathcal{T}_i$  the corresponding set of time indices.

The observation variance retains conjugacy and follows an inverse-gamma full conditional:

$$\sigma^2 \mid \cdot \sim \text{Inverse-Gamma}(\tilde{a}_\sigma, \tilde{b}_\sigma),$$

where

$$\tilde{a}_\sigma = a_\sigma + \frac{N}{2}, \quad \tilde{b}_\sigma = b_\sigma + \frac{1}{2} \sum_{i,t} (y_{it} - x_{it}^\top \beta - \alpha_i)^2,$$

and  $N$  denotes the total number of observations.

In contrast, the full conditional distribution of  $\tau$  does not belong to a standard family due to the half-Cauchy prior. Its conditional density is proportional to

$$p(\tau \mid \alpha) \propto \left[ \prod_{i=1}^J \mathcal{N}(\alpha_i \mid 0, \tau^2) \right] \pi(\tau), \quad \tau > 0.$$

These conditional distributions directly motivate a Metropolis-within-Gibbs sampler. At each iteration,  $\beta$ ,  $\alpha$ , and  $\sigma^2$  are updated through Gibbs steps from their full conditional distributions, while  $\tau$  is updated through a random-walk Metropolis-Hastings step. This hybrid scheme combines the computational convenience of conjugate updates with the flexibility required for the non-conjugate scale component.

### 3.3 Implementation details

Hyperparameters are selected to induce weakly informative priors. For the regression coefficients, the prior covariance matrix is set to  $B_0 = cI_p$ , with baseline choice  $c = 100$ , implying a large prior variance and only mild shrinkage toward zero. For the observation variance, the inverse-gamma hyperparameters are fixed at  $a_\sigma = 2$  and  $b_\sigma = 0.5$ , yielding a diffuse prior. The value  $b_\sigma = 0.5$  is chosen to be consistent with the empirical scale of the log-transformed outcome (whose sample variance is approximately 0.47), while remaining sufficiently weakly informative. The scale parameter of the half-Cauchy prior on  $\tau$  is set to  $A_\tau = 5$ , allowing for substantial between-station variability while still providing regularization. The robustness of posterior inference to alternative hyperparameter choices is examined through a dedicated sensitivity analysis reported below.

Posterior inference is performed using a Metropolis-within-Gibbs sampler run for 10,000 iterations. Based on the Heidelberger–Welch diagnostic, the first 1,000 iterations are discarded as burn-in. The remaining chain is thinned by retaining every fourth draw, yielding a final posterior sample of 2,250 draws.

The Metropolis step for  $\tau$  is implemented as a Gaussian random walk on  $\log \tau$ . The proposal standard deviation is calibrated by testing a range of candidate values and selecting one that yields a satisfactory acceptance rate, so as to avoid both excessively small proposal variances, which lead to poor exploration of the posterior, and excessively large variances, which would result in low acceptance and inefficient sampling. A standard deviation of 0.20 produces an acceptance rate of approximately 0.37, which lies within commonly recommended ranges. As results were robust to nearby choices, this specification was retained.

Finally, convergence is assessed using both graphical (trace plots and autocorrelation functions) and numerical diagnostics, including the Geweke statistic, the Heidelberger–Welch diagnostic, and the effective sample size.

### 3.3.1 Sensitivity to prior specification

To assess the robustness of posterior inference to prior calibration, a one-at-a-time sensitivity analysis is conducted. Starting from the baseline configuration  $c = 100$ ,  $A_\tau = 5$ , and  $b_\sigma = 0.5$ , each hyperparameter is varied separately while holding the others fixed. For computational efficiency, the sensitivity runs are based on shorter auxiliary chains of 5,000 iterations, with the first 200 discarded as burn-in, which are sufficient for assessing the stability of posterior summaries.

Sensitivity is evaluated with respect to the three prior components: the prior variance of the regression coefficients ( $c = 10, 1000$ ), the scale of the half-Cauchy prior on the station-level standard deviation ( $A_\tau = 2.5, 10$ ), and, the inverse-gamma scale parameter for the observation variance ( $b_\sigma = 1$ , with  $a_\sigma = 2$  fixed).

As reported in Table 1, posterior summaries are highly stable across all specifications. The posterior means of the intercept,  $\sigma^2$ , and  $\tau$  exhibit negligible variation, and acceptance rates for the Metropolis step on  $\tau$  remain within a satisfactory range.

**Table 1:** Sensitivity analysis under alternative prior specifications.

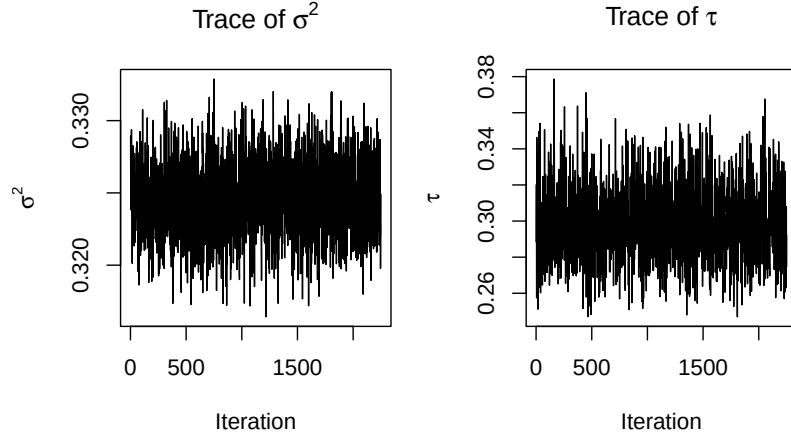
| Scenario                           | $c$  | $A_\tau$ | $b_\sigma$ | $\beta_0$ | $\sigma^2$ | $\tau$ | Acceptance ( $\tau$ ) |
|------------------------------------|------|----------|------------|-----------|------------|--------|-----------------------|
| Baseline                           | 100  | 5.0      | 0.5        | 2.603     | 0.325      | 0.299  | 0.388                 |
| Smaller prior variance for $\beta$ | 10   | 5.0      | 0.5        | 2.602     | 0.325      | 0.298  | 0.394                 |
| Larger prior variance for $\beta$  | 1000 | 5.0      | 0.5        | 2.603     | 0.325      | 0.299  | 0.401                 |
| Smaller $A_\tau$                   | 100  | 2.5      | 0.5        | 2.603     | 0.325      | 0.299  | 0.384                 |
| Larger $A_\tau$                    | 100  | 10.0     | 0.5        | 2.603     | 0.325      | 0.299  | 0.388                 |
| Larger $b_\sigma$                  | 100  | 5.0      | 1.0        | 2.603     | 0.325      | 0.298  | 0.393                 |

The results indicate that posterior inference is not materially driven by the specific prior calibration, suggesting that the likelihood is sufficiently informative relative to the prior.

## 4 Results

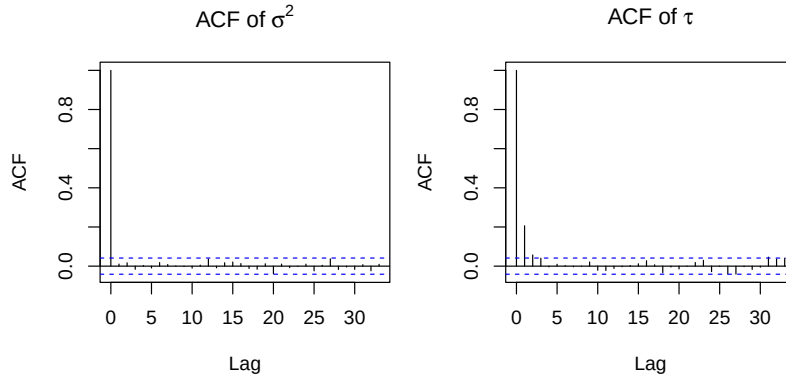
### 4.1 Convergence diagnostics

Trace plots of the post-burn-in chains for the variance and scale components (Figure 3) show no evidence of non-stationarity or systematic trends. Both chains fluctuate around stable levels, indicating satisfactory convergence of the Metropolis-within-Gibbs sampler.



**Figure 3:** Post-burn-in trace plots for the variance and scale components.

Autocorrelation functions for the variance and scale components are reported in Figure 4. For both  $\sigma^2$  and  $\tau$ , autocorrelation declines rapidly after the first lag and remains close to zero thereafter, indicating efficient mixing for both components.



**Figure 4:** Post-burn-in autocorrelation functions for the variance and scale components.

Autocorrelation functions were also inspected for representative regression coefficients and station-specific effects, selected according to their effective sample size in order to cover parameters with low, median, and high sampling efficiency. While some persistence is observed at the first lags for a subset of station-specific effects, autocorrelation decreases progressively and does not display long-range dependence.

This behaviour is consistent with the effective sample size diagnostics computed for all monitored parameters: ESS values range from 101 to 2040, with a mean of about 401. Regression coefficients



exhibit high sampling efficiency, with effective sample sizes typically ranging between about 1500 and 1600 for most parameters. The variance components also show very good mixing, with ESS values of about 1600 for  $\sigma^2$  and around 1200 for  $\tau$ , whereas lower ESS values are concentrated among a subset of station-specific effects, which is expected in hierarchical models and does not indicate pathological behavior.

The Geweke diagnostic provides a more conservative assessment of convergence. Regression coefficients and variance components have  $z$ -scores with absolute values below 1.96. For the station-specific effects, 27 out of 116 parameters fall outside this range. The Geweke  $z$ -scores for this block range from approximately 0.50 to 2.71, with exceedances occurring only on the positive side of the conventional threshold. These exceedances are concentrated among local random effects rather than structural parameters, and therefore suggest lower Monte Carlo stability for some station-level effects rather than a global failure of convergence.

The Heidelberger–Welch diagnostic was also used to assess stationarity and Monte Carlo precision. The stationarity test is passed by all monitored parameters, supporting the adequacy of the selected burn-in period. The half-width test is passed by 116 out of 136 parameters; the remaining failures are mainly concentrated among station-specific effects, indicating lower precision in estimating some local posterior means, consistently with the ESS and Geweke results.

These results are summarized in Table 2.

**Table 2:** Summary of Geweke and Heidelberger–Welch diagnostics by parameter block.

| Parameter block | Geweke min | Geweke max | HW stationarity | HW half-width |
|-----------------|------------|------------|-----------------|---------------|
| $\beta$         | -1.397     | 0.290      | 18/18           | 17/18         |
| $\alpha$        | 0.503      | 2.709      | 116/116         | 97/116        |
| $\sigma^2$      | 0.573      | 0.573      | 1/1             | 1/1           |
| $\tau$          | 1.412      | 1.412      | 1/1             | 1/1           |

Overall, the diagnostics indicate satisfactory convergence for the regression coefficients and variance components. Some station-specific effects display lower sampling efficiency and less stable Geweke statistics, but this behaviour is localized and consistent with the hierarchical structure of the model. Since the Heidelberger–Welch stationarity test is passed by all parameters and no persistent autocorrelation is observed, the posterior draws are considered adequate for the subsequent inferential analysis.

## 4.2 Posterior inference

The posterior estimates shown in Table 3 provide a clear characterization of the main temporal patterns in  $\text{PM}_{2.5}$  concentrations.

**Table 3:** Posterior means of coefficients, 95% HPD intervals, and approximate percentage effects relative to the baseline category.

| Variable             | Mean   | 95% HPD interval | Effect                   |
|----------------------|--------|------------------|--------------------------|
| Intercept (Jan, Mon) | 2.609  | [2.514, 2.712]   | Baseline ( $\sim 13.6$ ) |
| February             | -0.019 | [-0.1, 0.073]    | -1.8%                    |
| March                | -0.364 | [-0.448, -0.277] | -30.5%                   |
| April                | -0.617 | [-0.701, -0.531] | -46.0%                   |
| May                  | -0.744 | [-0.827, -0.655] | -52.5%                   |
| June                 | -0.122 | [-0.211, -0.037] | -11.5%                   |
| July                 | -0.600 | [-0.684, -0.515] | -45.1%                   |
| August               | -0.428 | [-0.516, -0.345] | -34.8%                   |
| September            | -0.802 | [-0.884, -0.714] | -55.2%                   |
| October              | -0.607 | [-0.69, -0.519]  | -45.5%                   |
| November             | -0.383 | [-0.468, -0.3]   | -31.8%                   |
| December             | -0.070 | [-0.149, 0.021]  | -6.8%                    |

The reported credible intervals are 95% highest posterior density (HPD) intervals, and effects whose intervals exclude zero are interpreted as credibly different from the baseline category.

The intercept corresponds to the expected value of the log-transformed outcome for the reference category (January and Monday), with a posterior mean of approximately 2.609, corresponding to a baseline  $\text{PM}_{2.5}$  level of about 13.6 on the original scale.

Posterior summaries for the variance components are reported in Table 4.

**Table 4:** Posterior summaries for variance components.

| Parameter  | Mean  | 95% HPD interval |
|------------|-------|------------------|
| $\sigma^2$ | 0.324 | [0.32, 0.33]     |
| $\sigma$   | 0.570 | [0.565, 0.574]   |
| $\tau$     | 0.298 | [0.263, 0.343]   |

The residual standard deviation exceeds the group-level standard deviation parameter  $\tau$ , indicating that within-station variability dominates persistent cross-station heterogeneity. At the same time, the group-level variation remains non-negligible, supporting the use of a hierarchical modeling specification.

#### 4.2.1 Temporal effects

The temporal coefficients reported in Table 3 reveal a clear seasonal pattern: relative to January, most months exhibit negative coefficients, indicating lower  $\text{PM}_{2.5}$  concentrations during the rest of the year.

The magnitude of these effects is substantively large. March, April, May, July, August, September, October, and November show clear reductions relative to January, with estimated decreases ranging from approximately 30% to more than 55%. The largest reduction is observed in September, with

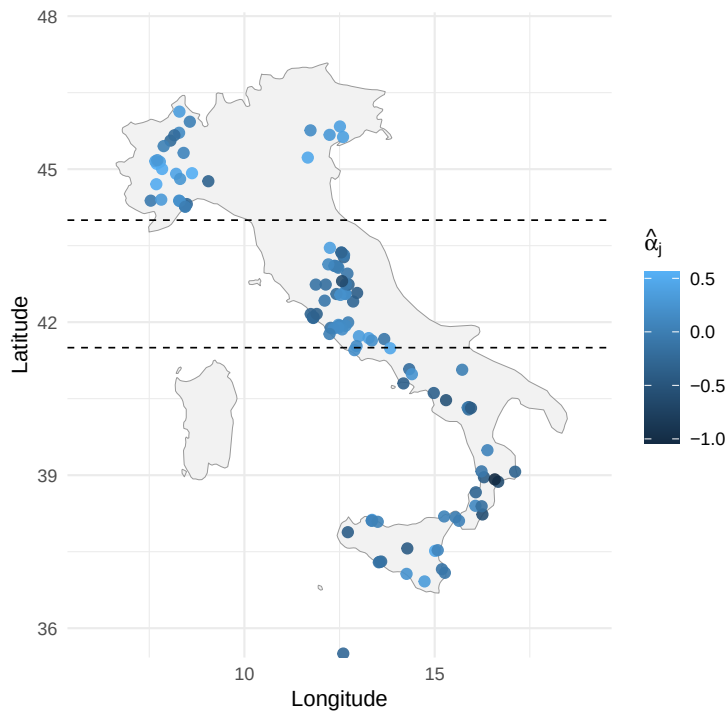
a posterior mean effect of about -55.2%. February and December are statistically indistinguishable from January, as their HPD intervals include zero. June also shows a smaller but credibly negative effect, corresponding to an estimated reduction of about -11.5%. This pattern is consistent with well-known environmental mechanisms, including increased heating emissions and less favorable atmospheric dispersion conditions during colder months.

For compactness, Table 3 reports only the intercept and monthly effects. The omitted day-of-week coefficients are generally small. Most of their 95% HPD intervals include zero, with the exception of Saturday, which shows a small positive effect relative to Monday, approximately equal to 2.6%. This suggests at most a weak day-of-week pattern once seasonal and station-level heterogeneity are accounted for.

#### 4.2.2 Spatial effects

Figure 5 displays the geographical distribution of monitoring stations, with colors representing the posterior mean of the station-specific random effects.

The analysis is exploratory and relies on latitude-based macro-area groups rather than on a formal spatial model. Specifically, macro-areas are defined using latitude thresholds at  $44^\circ$  and  $41.5^\circ$ . Positive values indicate structurally higher  $\text{PM}_{2.5}$  levels after controlling for temporal effects.



**Figure 5:** Spatial distribution of posterior mean station effects across Italy.

The spatial distribution reveals a clear North–South gradient in the station-level effects. Stations located in northern Italy tend to exhibit positive random effects, whereas southern stations display predominantly negative values. Central regions lie between these two patterns, with an average effect close to zero.

**Table 5:** Posterior summaries of average station effects by macro-area.

| Area           | Mean   | 95% HPD interval | Posterior probability |
|----------------|--------|------------------|-----------------------|
| North          | 0.191  | [0.141, 0.243]   | 1.0000000             |
| Center         | -0.019 | [-0.073, 0.029]  | 0.2244444             |
| South          | -0.117 | [-0.167, -0.065] | 0.0000000             |
| North - South  | 0.308  | [0.289, 0.326]   | 1.0000000             |
| North - Center | 0.210  | [0.194, 0.227]   | 1.0000000             |

This pattern is confirmed by a posterior comparison of average station effects across macro-areas (Table 5). The posterior mean station effect is positive in the North, approximately 0.19, with a 95% credible interval entirely above zero. By contrast, the South shows a negative average effect, approximately -0.12, with a 95% credible interval entirely below zero. The Center has an average effect close to zero, and its credible interval includes zero.

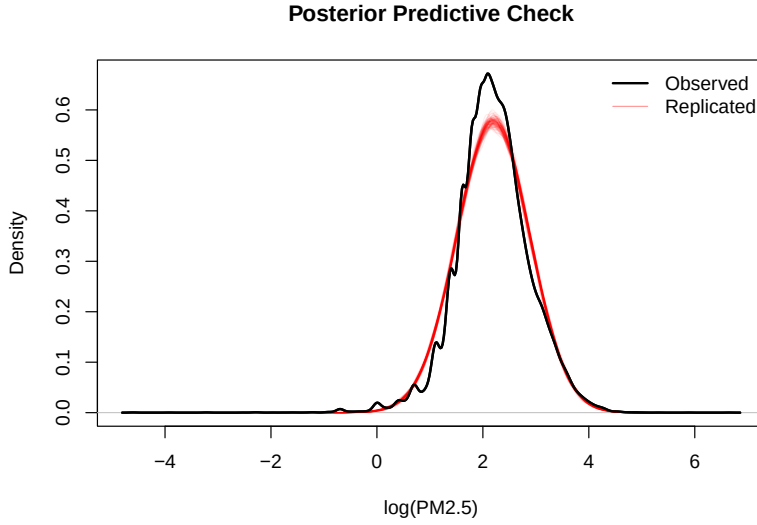
The North–Center contrast is also credibly positive, with an estimated difference of about 0.21 on the log scale. Overall, these results support the presence of a marked geographical gradient in station-level baseline pollution.

This evidence is consistent with well-documented geographical differences in pollution dynamics, including higher industrial intensity, traffic exposure, and less favorable atmospheric dispersion conditions in northern regions. At the same time, the dispersion of the station effects highlights substantial within-region heterogeneity, suggesting that local factors remain important determinants of air quality.

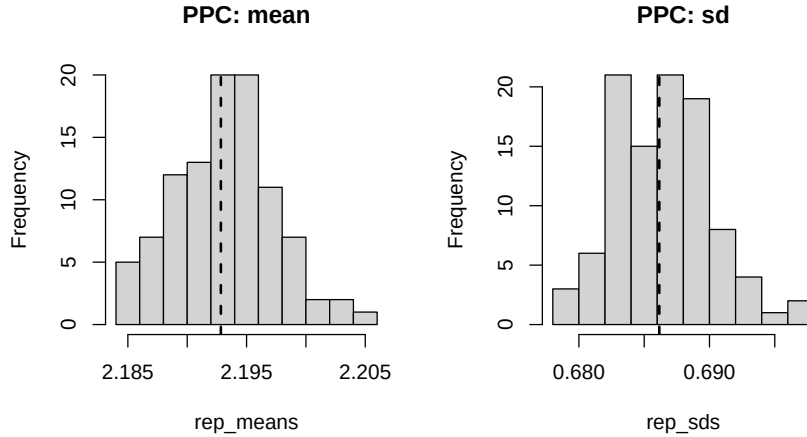
Overall, these results highlight the coexistence of broad geographical gradients and substantial local heterogeneity in  $\text{PM}_{2.5}$  dynamics.

### 4.3 Posterior predictive checks

Posterior predictive checks were performed to assess model adequacy. Replicated datasets were generated from the posterior distribution and compared to the observed data.



**Figure 6:** Posterior predictive density for the log-transformed  $\text{PM}_{2.5}$  outcome.



**Figure 7:** Posterior predictive checks based on summary statistics.

The posterior predictive summaries for the mean and standard deviation closely match the observed values (see Figures 6 and 7), with the observed statistics lying well within the distribution of replicated values. This suggests that the model provides an adequate representation of the first two moments of the data.

Furthermore, the posterior predictive density aligns reasonably well with the empirical distribution of log-transformed  $\text{PM}_{2.5}$ , although minor discrepancies are visible in the peak and tail regions. These differences are consistent with the Gaussian assumption and do not indicate major model misspecification.

Robustness checks based on alternative prior calibrations, reported in Section 3.3.1, confirm that the main posterior conclusions are stable.

## 5 Conclusion

This paper develops a Bayesian hierarchical model for daily  $\text{PM}_{2.5}$  concentrations, explicitly accounting for heterogeneity across monitoring stations and for the unbalanced structure of the panel. The Metropolis-within-Gibbs sampler shows satisfactory convergence according to several diagnostic tools, while sensitivity analysis indicates that posterior inference is robust to alternative prior specifications. Posterior predictive checks further suggest that the model captures the main features of the observed data reasonably well.

From a substantive perspective, the results reveal a pronounced seasonal pattern, with substantially higher  $\text{PM}_{2.5}$  levels during winter months. This finding reinforces the relevance of seasonally informed environmental policies, particularly those addressing emissions related to heating systems and increased urban activity. By contrast, there is little evidence of a strong weekly cycle: most day-of-week effects are small and not credibly different from zero, although Saturday shows a modest positive deviation. This suggests that short-term policy interventions (e.g., day-specific restrictions) are unlikely to be sufficient unless embedded within broader, structural measures.

The analysis also highlights persistent differences across monitoring stations and a clear North–South gradient in station-level effects. This pattern points to the importance of geographically differentiated policy approaches and suggests that local environmental conditions, urban structure, and regional economic activity jointly shape air pollution dynamics.

Despite these strengths, the analysis is subject to several limitations. First, the study is restricted to a single year, which prevents the investigation of longer-term trends, inter-annual variability, and the potential impact of policy interventions over time. In addition, the spatial coverage is limited to 116 monitoring stations and is not fully representative of the Italian territory, as some regions are underrepresented, potentially affecting the external validity of the results. Third, while the hierarchical structure partially mitigates data imbalance, irregular temporal coverage across stations may still influence inference for locations with substantial missing periods. Finally, the current specification does not explicitly model spatial dependence across stations, nor does it incorporate meteorological covariates (e.g., temperature, wind, humidity), which are known to play a key role in pollution dynamics.

These limitations suggest several directions for future research. Extending the dataset across multiple years and improving spatial coverage would strengthen the robustness of the empirical findings. Incorporating spatial correlation structures, such as Gaussian processes or conditional autoregressive (CAR) models, would allow for a more realistic representation of geographic dependence. Moreover, including meteorological and environmental covariates could improve both explanatory power and predictive performance. More flexible likelihood specifications, such as heavy-tailed or skewed distributions, may be better suited to capture extreme pollution episodes.

Overall, the results show that the hierarchical Bayesian framework provides a flexible and effective approach for the analysis of air pollution data, offering both methodological rigor and policy-relevant insights.